

## Chapter 1 – Introduction to SOC (Security Operations Center)

### What is SOC?

SOC (Security Operations Center) is a dedicated cybersecurity department that continuously monitors, detects, investigates, and responds to security incidents 24 hours a day, 7 days a week.

Think of SOC as the **security control room of an organization**, where security analysts constantly watch for cyber threats and attacks.

### Real-Life Example

Just like CCTV operators monitor cameras in a shopping mall 24/7, SOC analysts monitor an organization's:

- Network Traffic 
- User Activities 
- Firewalls 
- Servers 
- Endpoints 
- Security Alerts 

to detect and stop cyber attacks before they cause damage.

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## Major Responsibilities of a SOC Team

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### 1 Network Monitoring

#### ◆ What is Network Monitoring?

SOC analysts continuously monitor all incoming and outgoing network traffic to identify suspicious activities.

#### ◆ Why is it Important?

Attackers usually communicate with compromised systems through the network.

By monitoring traffic, SOC analysts can identify:

- Unauthorized access attempts
- Malware communication
- Data theft
- Suspicious IP addresses
- Unusual traffic patterns

#### 🌐 Example

If a company employee suddenly starts sending large amounts of data to an unknown foreign IP address, the SOC team will investigate immediately.

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### 2 Malware Analysis

#### ◆ What is Malware Analysis?

Malware analysis is the process of examining files, processes, and applications to determine whether they contain malicious code.

SOC analysts investigate:

- Running Processes
- Executable Files
- Applications

- Scripts
- Services

to check whether malware is present.

### Tools Used

One commonly used tool is:

- Sysinternals Suite

Using Sysinternals tools, analysts can examine:

- Running processes
- Registry entries
- Network connections
- Startup programs

### Example

If a computer suddenly becomes slow, analysts may discover a hidden malware process consuming system resources.

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## 3 Email Analysis

### ◆ What is Email Analysis?

Many cyber attacks begin through email.

SOC analysts examine emails to identify:

- Phishing attacks
- Malicious attachments
- Fake links
- Spoofed sender addresses

### ◆ Things Checked During Email Analysis

- ✓ Sender Email Address
- ✓ Sender IP Address

- ✓ Email Headers
- ✓ Attachments
- ✓ URLs inside the email

### Example

An employee receives an email claiming to be from a bank asking for login credentials. SOC analysts investigate whether the email is legitimate or a phishing attempt.

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### Threat Intelligence

#### ◆ What is Threat Intelligence?

Threat Intelligence is the process of collecting information about current and future cyber threats.

The goal is to understand:

- How attackers operate
  - What attacks may occur
  - How to defend against them
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### IOC (Indicators of Compromise)

When a cyber attack occurs, analysts collect evidence known as Indicators of Compromise (IOCs).

#### Common IOCs

- Malicious IP Addresses
- Domain Names
- Email Addresses
- File Hashes (MD5, SHA1, SHA256)
- URLs
- Registry Keys

### Purpose

These indicators help identify infected systems and prevent future attacks.

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### Updating Security Tools

After collecting IOCs, organizations update their security solutions:

- Antivirus (AV)
- Firewall (FW)
- IDS (Intrusion Detection System)
- IPS (Intrusion Prevention System)
- SIEM
- EDR

to detect similar attacks automatically in the future.

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### Threat Hunting

#### What is Threat Hunting?

Threat Hunting is the proactive process of searching for hidden threats inside an organization's environment using Threat Intelligence data.

#### Process

1. Collect IOC Data
2. Analyze Threat Information
3. Search Entire Environment
4. Identify Infected Systems
5. Remove Threats

### Career Opportunity

The Threat Researcher and Threat Hunter roles are highly demanded in cybersecurity.

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### 5 Vulnerability Assessment

#### ◆ What is Vulnerability Assessment?

Vulnerability Assessment is the process of identifying weaknesses in IT infrastructure before attackers exploit them.

#### Assets Checked

- Firewalls
- Routers
- Switches
- Servers
- Client Systems
- Applications
- Operating Systems

#### Analysts Check

- ✓ Software Versions
- ✓ Open Ports
- ✓ Running Services
- ✓ Security Configurations
- ✓ Missing Updates

#### 🌐 Example

If a server is running an outdated version of Windows with a known vulnerability, attackers may exploit it to gain access.

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### 6 Digital Forensics

#### ◆ What is Digital Forensics?

Digital Forensics is the investigation process performed after a cyber attack to determine:

- What happened?
- How did it happen?
- When did it happen?
- Who performed the attack?
- Which systems were affected?

### Real-Life Example

Similar to how police investigate a crime scene, digital forensic investigators analyze digital evidence after a cyber attack.

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## Types of Attack Sources

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### User-Based Attacks

These attacks occur because of user actions.

#### Examples

- Downloading malicious software
- Installing fake browser extensions
- Running infected files
- Opening malicious email attachments

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### Network-Based Attacks

These attacks exploit network weaknesses.

#### Examples

- Open Ports
- Weak RDP Credentials
- SMB Vulnerabilities
- Misconfigured Services

## Two Common Ways Attackers Gain Access

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### 1 Password-Based Attacks

The attacker gains access by:

- Guessing passwords
- Brute Force Attacks
- Credential Stuffing
- Password Spraying

#### Example

Using weak passwords such as:

- ✗ admin123
- ✗ password123
- ✗ welcome123

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### 2 Vulnerability-Based Attacks

Attackers exploit software weaknesses using exploits.

#### Example

An SMB vulnerability allows remote code execution on a server.

The attacker uses an exploit to gain access without knowing any password.

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### Social Engineering Attacks

#### ◆ What is Social Engineering?

Social Engineering is the manipulation of people into performing actions that compromise security.

Attackers target humans instead of systems.



## Common Social Engineering Techniques

### 1 Malicious Link Attack

The attacker sends a fake link.

The victim clicks it and becomes infected.

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### 2 Fake Website Attack

The attacker creates a website that looks identical to a real website.

The victim enters credentials, which are stolen.

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### 3 Malicious Email Attachment

The attacker sends an email containing:

- Malware
  - Documents with macros
  - Scripts
  - Trojans
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### 4 Infected USB/Pen Drive Attack

The attacker leaves infected devices in public places.

Examples:

- USB Drives
- CDs/DVDs
- Memory Cards

Curious users plug them into their systems, causing malware execution.

### Example

An employee finds a USB drive in the parking lot and connects it to the company computer.

The malware automatically infects the system.

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### **5** USB Charger Attack (Juice Jacking Concept)

A modified charging station or cable can be used to transfer malicious data.

#### Requirements

- USB Data Connection
- USB Debugging Enabled
- Developer Mode Enabled

#### Note

Modern smartphones have stronger protections, but unsecured devices may still be vulnerable.

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### Role of Digital Forensics in Attack Investigation

When an attack occurs, digital forensic investigators determine:

- Attack Method
- Entry Point
- Timeline
- Impact
- Evidence

Their primary objective is to discover exactly how the compromise occurred.

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## Scenario: Proving a User Caused the Infection

### Situation

A malware infection occurs.

The user claims:

"I did not download anything."

"I was not even present when the attack happened."

The forensic investigator must prove what actually happened using evidence.

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### Evidence Collection Process

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#### 1 Event Viewer Investigation

#### Check Security Logs

Investigators review:

- Successful Logins
- Failed Logins

#### Analysis

##### Scenario A

Only successful login present.

✓ Likely legitimate user login.

##### Scenario B

Many failed logins followed by a successful login.

⚠ Possible password attack or brute force attempt.

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### **2 Browser Investigation**

Chrome stores user activity information.

#### **Common Location**

AppData\Local\Google\Chrome\User Data

Investigators check:

- Download History
  - Browsing History
  - Saved Files
  - Extensions
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### **3 Application Logs**

Application logs reveal:

- Software Installations
- Program Execution
- Updates
- Errors

This helps identify when malicious software was installed.

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### **4 TEMP Folder Investigation**

#### **Common Location**

%temp%

Investigators check suspicious files and examine:

- Created Date
- Modified Date
- Accessed Date

These timestamps help build the attack timeline.



## Disk Imaging and Memory Capture

Digital forensic investigations should never directly modify original evidence.

Therefore investigators create:

### Hard Disk Clone (Disk Image)

A bit-by-bit copy of the original hard drive.

This preserves evidence integrity.

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### RAM Capture (Memory Dump)

Captures the contents of RAM.

Useful for identifying:

- Running Processes
- Malware in Memory
- Active Sessions
- Network Connections
- Encryption Keys



### Example

A malware process may disappear after shutdown, but memory capture can preserve evidence before the system is powered off.

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### Important Interview Question

**Q: What are the main functions of a SOC Team?**

**Answer:**

1. Network Monitoring
  2. Malware Analysis
  3. Email Analysis
  4. Threat Intelligence
  5. Threat Hunting
  6. Vulnerability Assessment
  7. Incident Response
  8. Digital Forensics
  9. Security Monitoring
  10. Log Analysis
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### Chapter 1 Summary

A SOC (Security Operations Center) is a 24×7 cybersecurity department responsible for monitoring, detecting, investigating, and responding to cyber threats. SOC analysts perform network monitoring, malware analysis, email analysis, threat intelligence, threat hunting, vulnerability assessment, and digital forensics. Most attacks generally occur through weak passwords, software vulnerabilities, or social engineering techniques, making continuous monitoring and investigation critical for organizational security.

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